

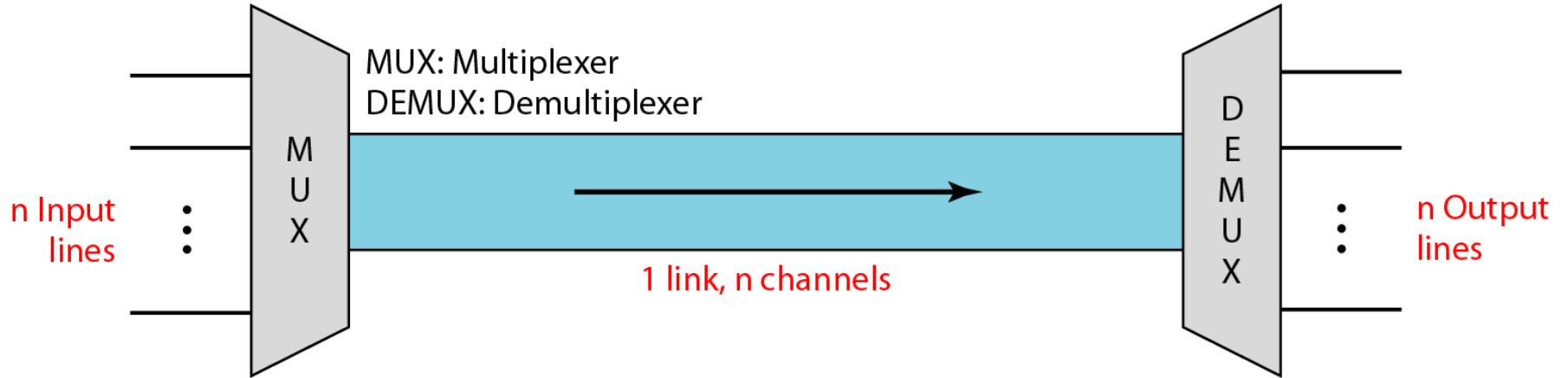
MULTIPLEXING

Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared. Multiplexing is the set of techniques that allows the (simultaneous) transmission of multiple signals across a single data link. As data and telecommunications use increases, so does traffic.

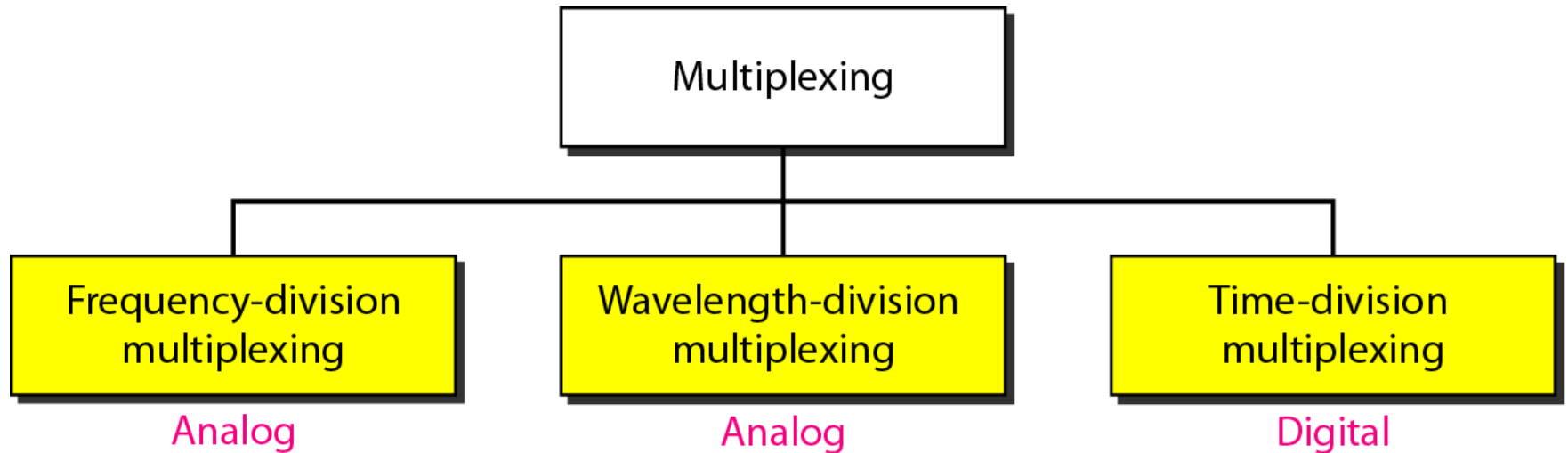
Topics discussed in this section:

- ☐ **Frequency-Division Multiplexing**
- ☐ **Wavelength-Division Multiplexing**
- ☐ **Synchronous Time-Division Multiplexing**
- ☐ **Statistical Time-Division Multiplexing**

Dividing a link into channels

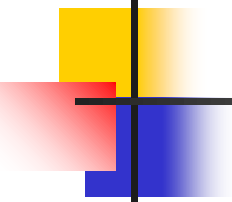


Categories of multiplexing



Frequency-division multiplexing (FDM)

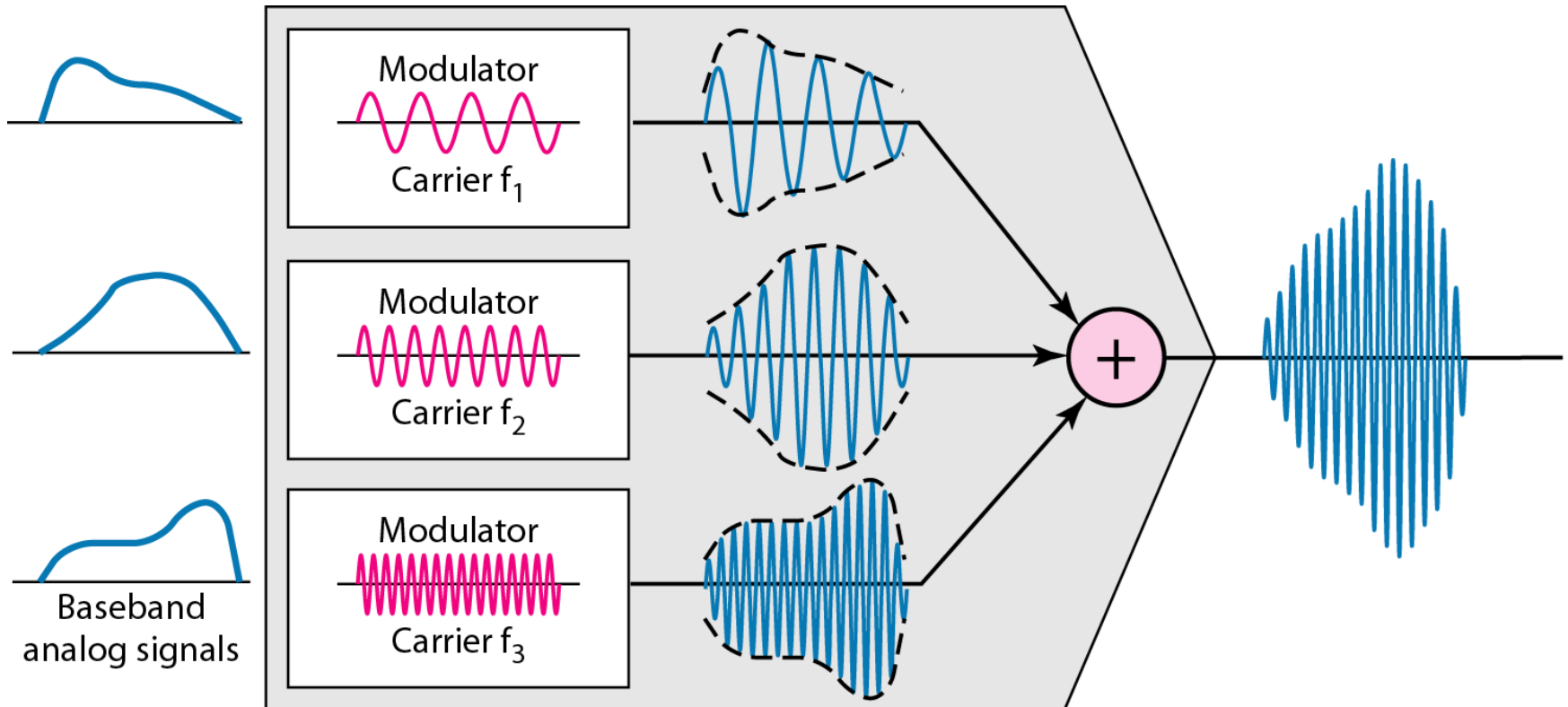




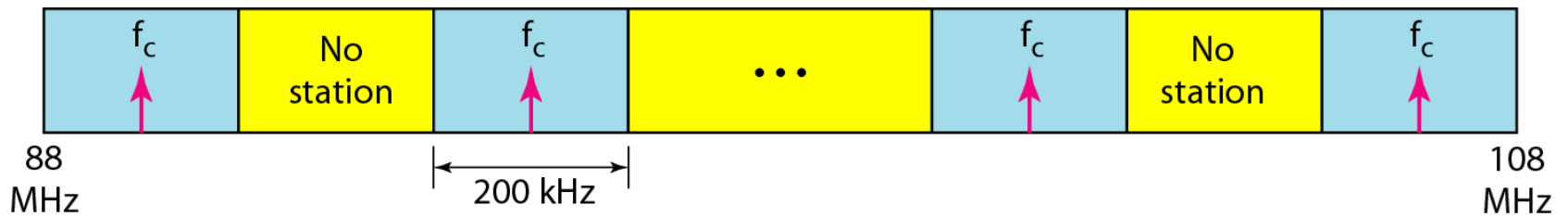
Note

FDM is an analog multiplexing technique that combines analog signals.

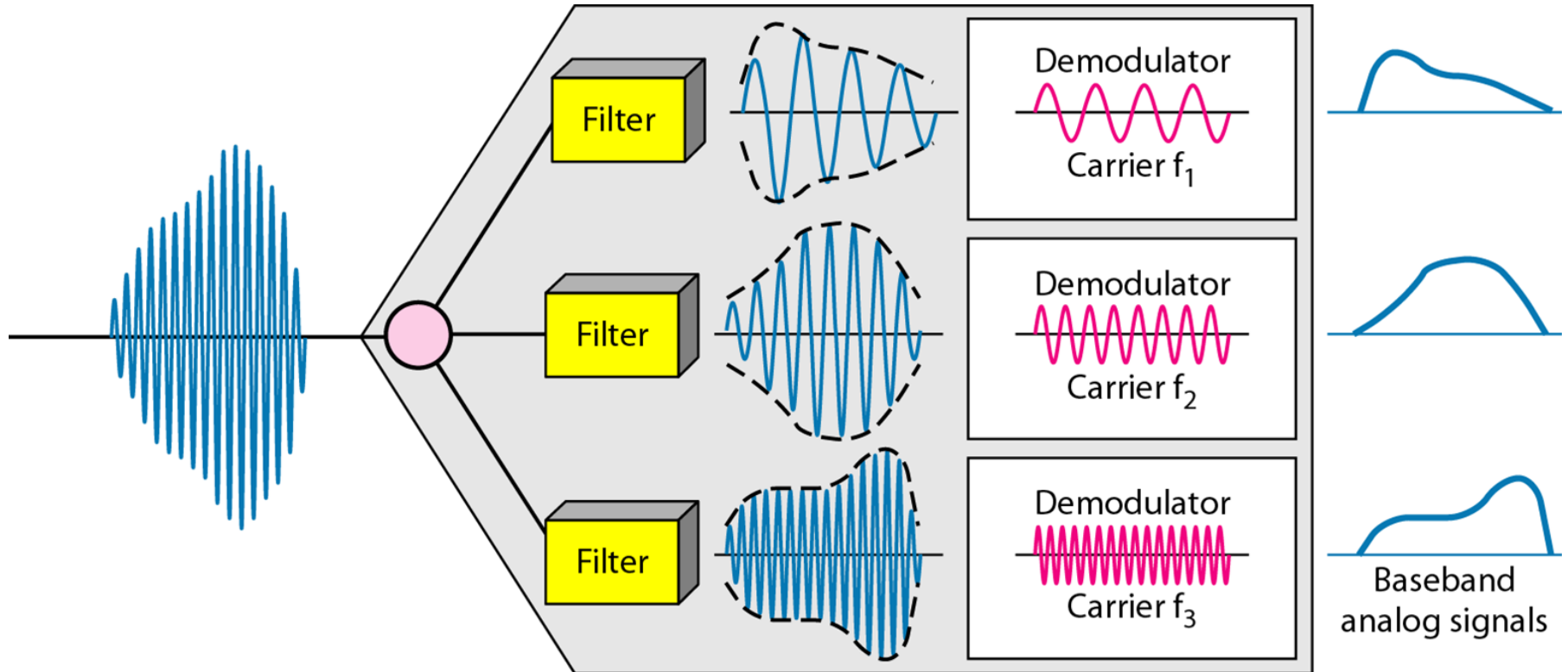
FDM process



FM



FDM demultiplexing example





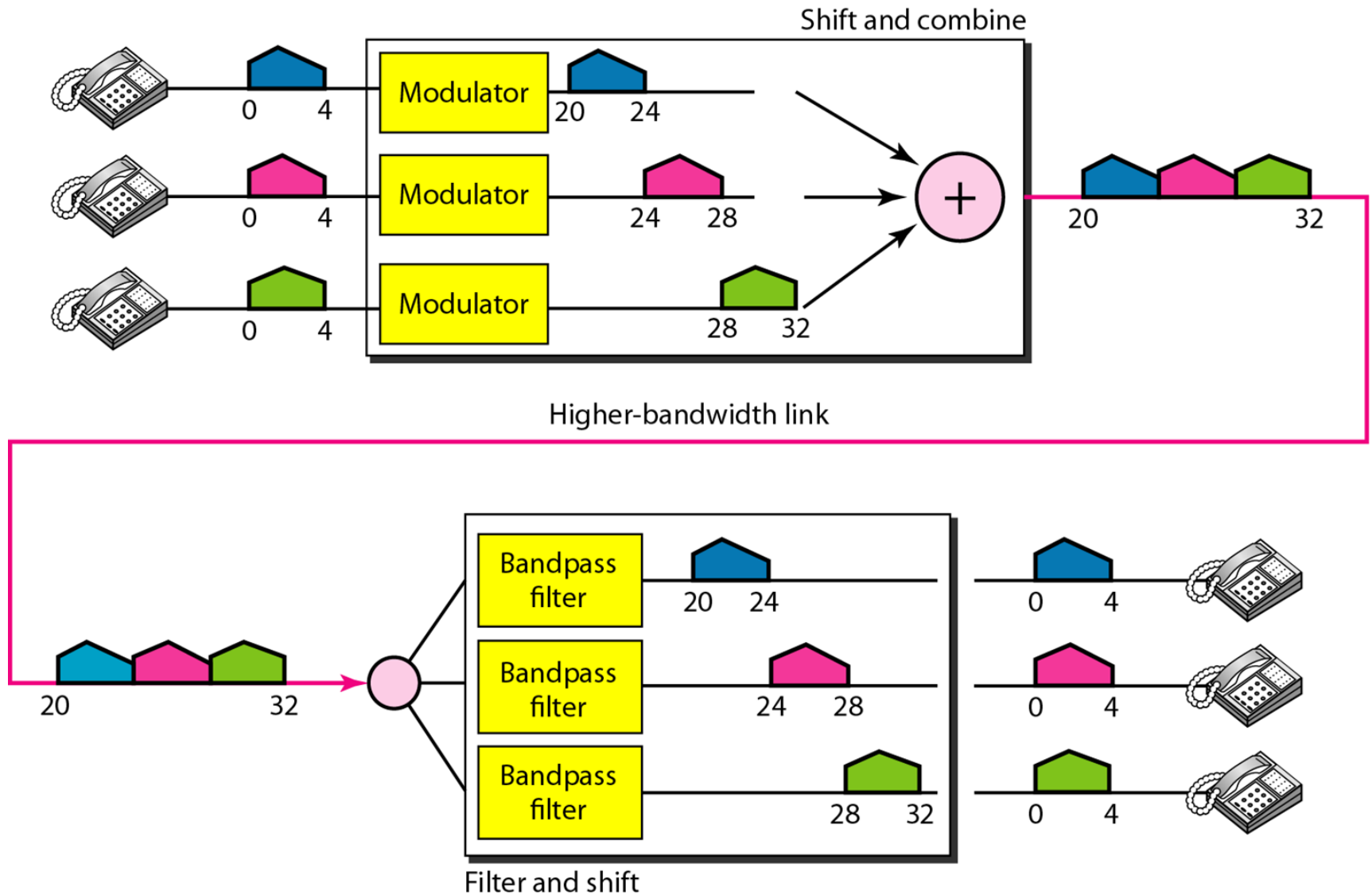
Example 1

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

Solution

We shift (modulate) each of the three voice channels to a different bandwidth, as shown in Figure 6.6. We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them as shown in Figure 1.

Figure 1: Example 1





Example 2

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

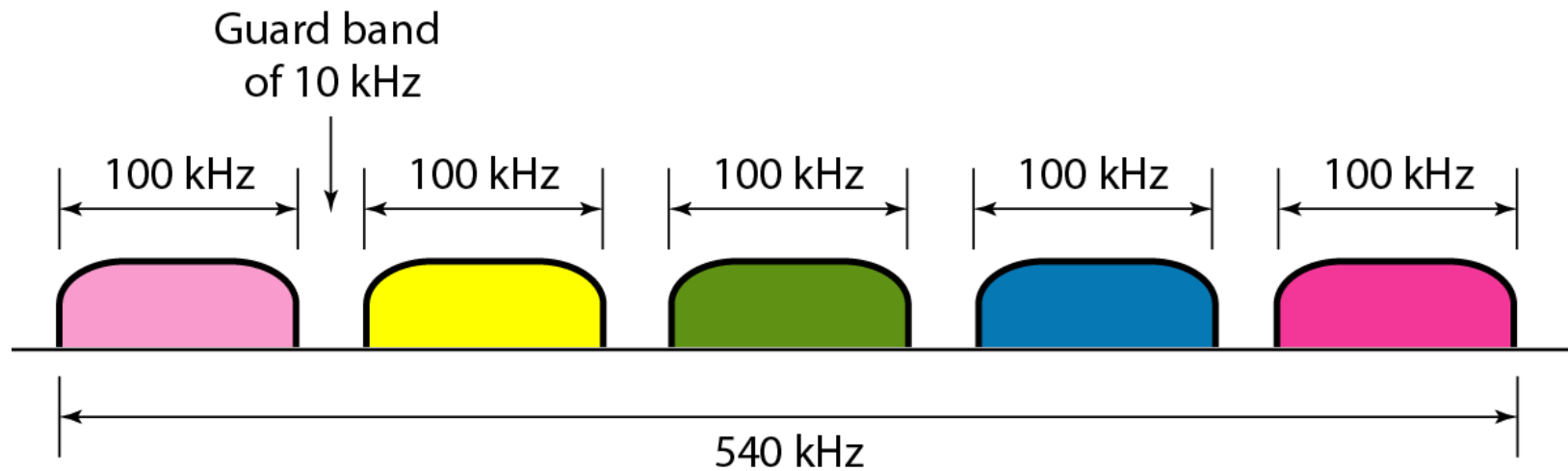
Solution

For five channels, we need at least four guard bands. This means that the required bandwidth is at least

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$

as shown in Figure 2.

Figure 2 *Example 2*



Example 3

Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel of 1 MHz. Design an appropriate configuration, using FDM.

Solution

The satellite channel is analog. We divide it into four channels, each channel having $1\text{M}/4=250\text{-kHz}$ bandwidth.

Each digital channel of 1 Mbps must be transmitted over a 250KHz channel. Assuming no noise we can use Nyquist to get:

$C = 1\text{Mbps} = 2 \times 250\text{K} \times \log_2 L \rightarrow L = 4$ or $n = 2$ bits/signal element.

One solution is 4-QAM modulation. In Figure 3 we show a possible configuration with $L = 16$.

Figure 3 *Example 3*

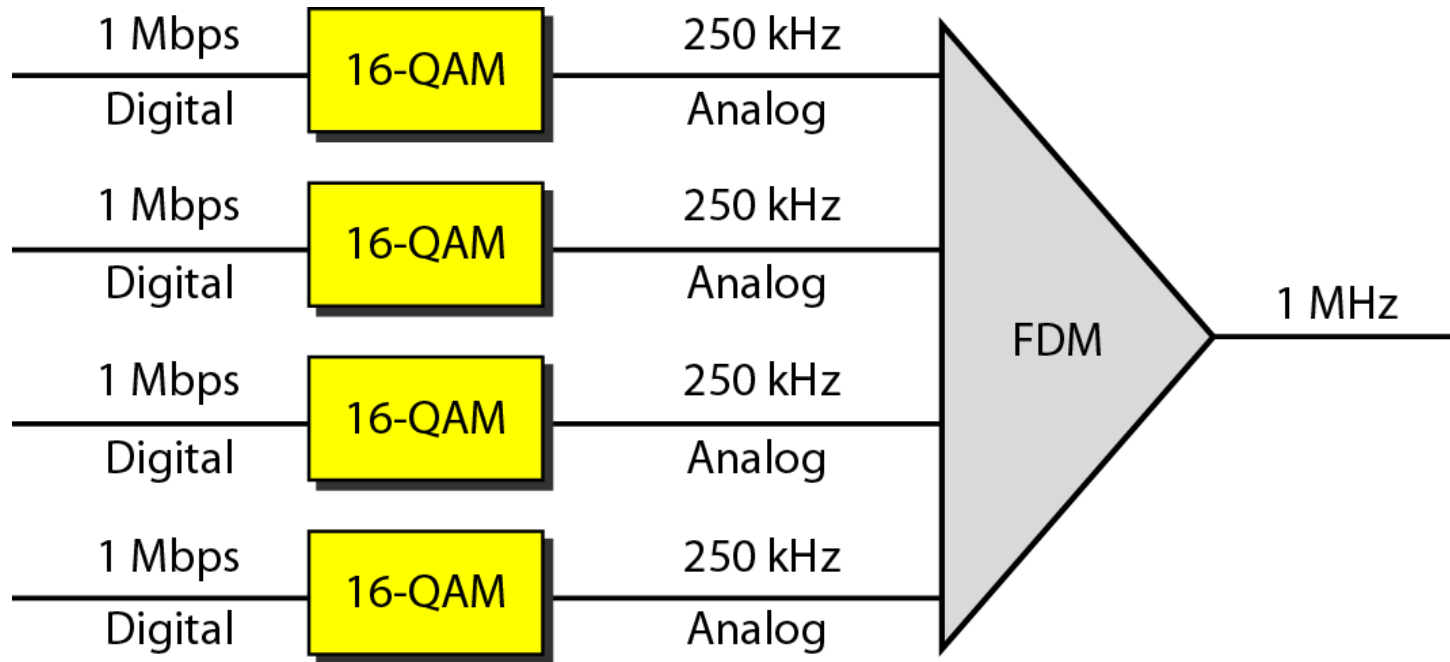
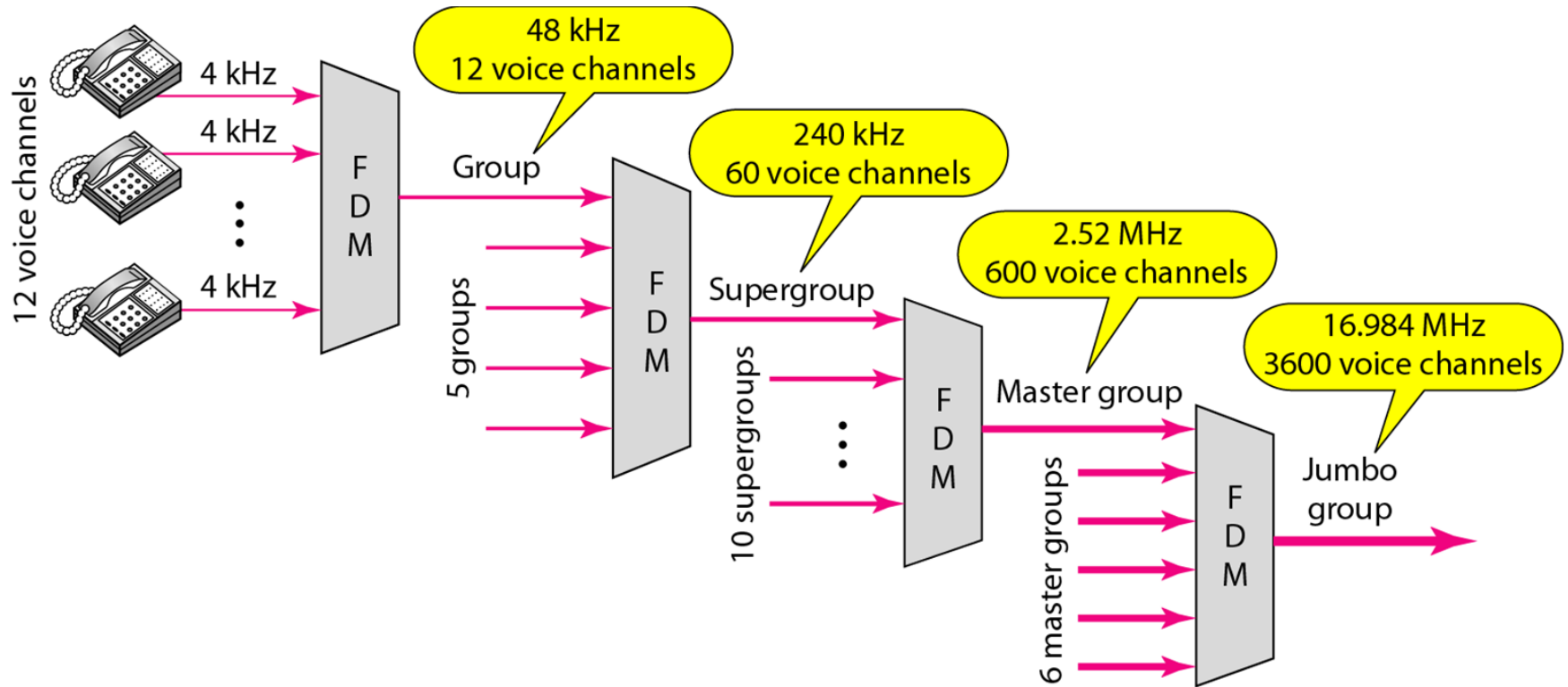


Figure 4 *Analog hierarchy (North American Hierarchy)*





Example 4

The Advanced Mobile Phone System (AMPS) uses two bands. The first band of 824 to 849 MHz is used for sending, and 869 to 894 MHz is used for receiving. Each user has a bandwidth of 30 kHz in each direction. How many people can use their cellular phones simultaneously?

Solution

Each band is 25 MHz. If we divide 25 MHz by 30 kHz, we get 833.33. In reality, the band is divided into 832 channels. Of these, 42 channels are used for control, which means only 790 channels are available for cellular phone users.

Figure 5 Time Division Multiplexing (TDM)

